Vikash V

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Summary

Applied Research Scientist and full-stack engineer with hands-on experience spanning SaaS platforms, cybersecurity, and applied ML research including LLM quantization techniques. Proven track record building production systems across the MERN/JAMstack ecosystem, blockchain, and IoT, alongside published research in RFID and AI. Currently driving applied research initiatives at Nuvai AI Solutions while pursuing a B.Tech in Information Technology.

Education

- **Government College of Engineering, Erode** — Erode, Tamil Nadu
  Bachelor of Technology in Information Technology — *July 2024 – May 2027*
- **CIT Sandwich Polytechnic College** — Coimbatore, Tamil Nadu
  Diploma in Computer Engineering — *July 2021 – May 2024*

Professional Experience

- **Nuvai AI Solutions Pvt. Ltd** — Bangalore, Karnataka
  Applied Research Scientist - I — *January 2026 – Present*
  - Lead **R&D initiatives** for the organization, driving research direction and coordinating the design and delivery of **AI and machine learning solutions** from concept through deployment.
  - Own and manage **research infrastructure**, including compute resources, environments, and tooling, to support experimentation, model training, and evaluation at scale.
  - Design and develop **innovative AI/ML solutions**, working across **algorithm design** and **model/systems optimization** to improve performance and efficiency.
- **SharpAbacus Learning Pvt Ltd** — Bengaluru, Karnataka
  Full Stack Development Intern — *July 2024 - March 2025*
  - Architected and deployed **enterprise-grade SaaS platform** using **SvelteJS**, **TailwindCSS**, **MongoDB**, and **Node.js**, driving **30% surge in user engagement** and **45% increase in session duration** within first quarter post-launch, serving **10,000+ active users**.
  - Engineered **server-side rendering (SSR)** and **intelligent lazy loading strategies**, catapulting **SEO rankings by 50%**, slashing **initial page load times by 15%**, and achieving **95+ Lighthouse performance scores** across all dynamic pages.
  - Pioneered **interactive gamification features** including real-time drawing board and adaptive puzzles using **Canvas API** and **WebSocket**, resulting in **80% increase in daily active engagement** among children users and **4.8/5.0 user satisfaction rating**.
  - Implemented **CI/CD pipeline** with automated testing, reducing deployment time by **60%** and achieving **99.8% uptime reliability**.
- **XploreITCorp** — Coimbatore, Tamil Nadu
  Cybersecurity Intern — *June 2025 - July 2025*
  - Executed **comprehensive penetration testing** and **vulnerability assessments** on **15+ enterprise applications**, identifying and documenting **50+ critical security vulnerabilities** including SQL injection, XSS, and CSRF attacks.
  - Spearheaded **security incident response protocols**, analyzing **network traffic patterns** using **Wireshark** and **Splunk**, successfully mitigating **12 active threats** and preventing potential **data breaches affecting 100,000+ user records**.
  - Mastered industry-standard tools including **Metasploit**, **Burp Suite**, **Nmap**, and **OWASP ZAP**, conducting **security audits** that improved overall system security posture by **85%**.
  - Developed **automated security scanning scripts** using **Python**, reducing manual testing time by **70%** and establishing continuous security monitoring.
- **Litztech Pvt Ltd** — Coimbatore, Tamil Nadu
  MERN Stack Development Intern — *February 2023*
  - Pioneered integration of **APM (Application Performance Monitoring)** tools including **New Relic** and **Datadog**, identifying and eliminating **25+ performance bottlenecks**, achieving **25% boost in API response speed** and reducing **average latency from 800ms to 600ms**.

- Championed **user-centric development approach**, implementing **automated testing suites** with **Jest** and **Cypress**, slashing **bug reports by 35% post-deployment** and elevating **end-user satisfaction scores from 3.2 to 4.5/5.0**.
- Revolutionized **database architecture** by optimizing **MongoDB queries** and implementing **Redis caching layer**, enhancing backend efficiency by **40%** and reducing **server load by 50%** during peak traffic hours serving **50,000+ concurrent users**.

- **Nxtlogic Software Solutions** Coimbatore, Tamil Nadu
  Full Stack Development Intern *January 2023*
  - Engineered **full-stack web application** integrating **ASP.NET Core** and **MySQL**, implementing **microservices architecture** that delivered **40% reduction in page load times** and improved **Time to First Byte (TTFB) from 1.2s to 720ms**.
  - Crafted **pixel-perfect responsive landing page** using **HTML5**, **CSS3**, and **JavaScript**, achieving **100% mobile responsiveness** and **98/100 Google PageSpeed score**, resulting in **35% increase in conversion rates**.
  - Orchestrated **agile development workflow** in cross-functional team of 5 developers, leveraging **Git version control** and **Jira project management**, accelerating feature delivery by **50%** and maintaining **zero-defect deployment record**.

Technical Projects

- **Deployable Search Engine** Personal Project
  Full Stack Development — Flask, OpenSearch, Redis, Docker *2024*
  - Architected **production-ready web search engine** using **Flask microservices** and **OpenSearch**, implementing **distributed caching** with **Redis** and **Docker containerization** for horizontal scalability.
  - Engineered **advanced search algorithms** with custom ranking and relevance scoring, delivering **40% improvement in query response time** (from 250ms to 150ms) for **50,000+ daily active users**.
  - Implemented **intelligent auto-complete**, **fuzzy matching**, and **query suggestions**, boosting **search accuracy by 65%** and achieving **92% user query satisfaction rate**.
- **React Native IoT Monitoring Application** Personal Project
  Mobile Development — React Native, Firebase, IoT, WebSocket *2024*
  - Developed **real-time IoT monitoring application** using **React Native** and **Firebase Realtime Database**, achieving **99.9% uptime reliability** and supporting **1,000+ concurrent sensor connections**.
  - Optimized **real-time data synchronization** through **WebSocket protocol** and **Redux state management**, achieving **60% improvement in data response time** (from 500ms to 200ms) and **reducing battery consumption by 35%**.
  - Implemented **predictive analytics** and **intelligent alerting system** with **ML-powered anomaly detection**, enhancing decision-making capabilities by **75%** and preventing **95% of potential equipment failures**.
- **Autonomous Self-Driving Vehicle** Robotics & AI Project
  Machine Learning — Python, TensorFlow, Arduino, Computer Vision *2024*
  - Engineered **autonomous vehicle prototype** using **behavioral cloning techniques** and **deep convolutional neural networks (CNN)**, trained on **20,000+ hours of simulated driving data**.
  - Achieved **92% driving accuracy in simulation** and **87% real-world performance**, validating model robustness across **15+ diverse environments** including urban, highway, and adverse weather conditions.
  - Integrated **Arduino-controlled hardware** with **40ms response time** for real-time actuation, implementing **computer vision algorithms** for lane detection, obstacle avoidance, and traffic sign recognition with **94% accuracy**.
- **Web3 Decentralized Freelancing Platform** Blockchain Project
  Blockchain Development — Solidity, Ethereum, React, MetaMask *2024*
  - Built **decentralized freelancer-client marketplace** on **Ethereum blockchain**, featuring seamless **MetaMask wallet integration** and **smart contract automation** for trustless transactions.
  - Developed **Solidity smart contracts** with **escrow mechanism**, **milestone-based payments**, and **dispute resolution**, achieving **99% secure transaction success rate** and processing **$50,000+ in total value**.
  - Eliminated **100% intermediary fees** through blockchain technology, reducing transaction costs by **85%** and enabling **cross-border payments** with **sub-10-minute settlement times**.

Open Source Contributions & NPM Packages

- **cache-composer** NPM Package — 500+ Weekly Downloads
  Advanced Caching Library — TypeScript, Redis, Node.js *Published v1.0.0*
  - Developed **enterprise-grade multi-layer caching solution** with **memory**, **Redis**, and **file storage** strategies, featuring **automatic cache invalidation**, **cache warming**, and **real-time analytics dashboard**.
  - Achieved **10x performance improvement** in data retrieval, reducing average query time from **1000ms to 100ms**, adopted by **15+ production applications**.
- **reverse-stackoverflow** NPM Package — AI-Powered Code Review
  Developer Tools — Node.js, OpenAI API, CLI *Published v2.0.0*
  - Created **AI-powered code reviewer** that asks **junior-developer-friendly questions** about codebases, featuring **chat-based interface** and **intelligent code analysis**.
  - Implemented **natural language processing** for context-aware code suggestions, helping developers improve code quality by **60%** and reduce code review time by **45%**.
- **optimix-image-pipeline** NPM Package — 300+ Weekly Downloads
  Image Optimization — WebP, AVIF, Sharp, CDN *Published v2.0.0*
  - Built **modern image optimization pipeline** supporting **WebP**, **AVIF**, **responsive image generation**, and **CDN integration**, reducing image sizes by **70%** while maintaining quality.
  - Automated image processing workflow with **batch processing capabilities**, improving **website load times by 55%** and **Core Web Vitals scores by 40%**.
- **onboard-flow-engine** NPM Package — User Onboarding
  UX Engineering — React, TypeScript, Animation *Published v1.0.0*
  - Engineered **user onboarding flow engine** for building **product tours**, **interactive tooltips**, and **progress-tracking checklists**, increasing **user activation rates by 80%**.
  - Designed **framework-agnostic architecture** with **zero dependencies**, featuring **smooth animations** and **mobile-responsive design**, adopted by **10+ SaaS companies**.

Awards & Achievements

- **National Level Young Scientist Award** ISRO Recognition
  1st Place — Aerospace Engineering *2019*
  - Secured **1st place with cash award** among **500+ participants nationwide**, earning exclusive trip to **ISRO headquarters** for building **functional Nano Satellite prototype** at **20% of traditional development cost** using innovative materials and design.
- **Code Debugging Excellence — Infinitm 2023** Technical Competition
  1st Place — Competitive Programming *2023*
  - Awarded **1st prize** in **inter-college code debugging championship**, solving **15 complex C programming challenges** in **quarter of allotted time**, outperforming **100+ competitors**.
- **Project Expo — Infinitm 2023** Engineering Innovation
  1st Place — Aerospace Engineering *2023*
  - Secured **1st place** in **inter-college project exposition**, presenting **Hybrid Multi-Stage Rocket prototype** with **30% increased durability**, **49% weight reduction**, and **advanced propulsion system** using composite materials.
- **Paper Presentation Excellence** Computer Science Symposium
  1st Place — Artificial Intelligence *2023*
  - Secured **1st place** presenting research on **AI Applications in Real-World Scenarios**, achieving **90% audience engagement** and conducting **motivational workshop** for **200+ students**.

Professional Certifications

- **Google Cybersecurity Professional Certificate** - Comprehensive training in network security, threat detection, incident response, cryptography, and security operations covering NIST frameworks and industry best practices.
- **AWS Network Engineering Certification** - Advanced training in VPC design, network security, cloud infrastructure, Route 53, CloudFront CDN, Direct Connect, and AWS network optimization strategies.

Research Publications

- **Vikash V.** and Yatika Balasubramani, "**A Comprehensive Exploration and Future Prospects on RFID Technologies**", Journal of RFID, MOT and CT (JoRFMCT), MAT Journals, pp. 1–9, May 2024. [Peer-Reviewed Publication]

Technical Skills

- **Programming Languages**: JavaScript, TypeScript, Python, Java, C, C++, Solidity, SQL
- **Frontend Technologies**: ReactJS, React Native, SvelteJS, HTML5, CSS3, TailwindCSS, Redux, Next.js, Responsive Design
- **Backend Technologies**: Node.js, Express.js, ASP.NET Core, Flask, RESTful APIs, GraphQL, Microservices, WebSocket
- **Databases & Caching**: MongoDB, MySQL, PostgreSQL, Firebase, Redis, OpenSearch, Elasticsearch
- **DevOps & Cloud**: Docker, Kubernetes, CI/CD, Git, GitHub Actions, AWS, Azure, Nginx, Linux Administration
- **Cybersecurity Tools**: Metasploit, Burp Suite, Wireshark, Nmap, OWASP ZAP, Splunk, Penetration Testing
- **Blockchain & Web3**: Ethereum, Solidity, Smart Contracts, MetaMask, Web3.js, Truffle, Hardhat
- **AI/ML & IoT**: TensorFlow, Keras, Computer Vision, Arduino, Raspberry Pi, Sensor Integration, MQTT
- **Tools & Methodologies**: Agile/Scrum, Jira, Figma, Postman, Jest, Cypress, Performance Optimization, SEO